Закаблукова
Анастасия
ИЗТ-1.1.
2 курс.

$$\begin{cases} 1\cdot x_1 + 2\cdot x_2 + 3\cdot x_3 + 5\cdot x_4 = 11 \\ 2\cdot x_1 + 1\cdot x_2 + 3x_3 + 5\cdot x_4 = 11 \\ 4\cdot x_1 + 1\cdot x_2 + 3x_3 + 5x_4 = 13 \\ 7x_1 + 5\cdot x_2 + 3x_3 + 1\cdot x_4 = 16 \end{cases}$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 3 & 5 & 11 \\ 2 & 1 & 3 & 5 & 11 \\ 4 & 1 & 3 & 5 & 13 \\ 7 & 5 & 3 & 1 & 16 \end{array}\right) \begin{array}{l} \\ \overline{II} - 2\overline{I} \\ \overline{III} - 4\overline{I} \\ \overline{IV} - 7\overline{I} \end{array} \sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & 5 & 11 \\ 0 & -3 & -3 & -5 & -11 \\ 0 & -7 & -9 & -15 & -31 \\ 0 & -9 & -18 & -34 & -61 \end{array}\right) \overline{II} : (-3) \sim$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & 5 & 11 \\ 0 & 1 & 1 & 5/3 & 11/3 \\ 0 & -7 & -9 & -15 & -31 \\ 0 & -9 & -18 & -34 & -61 \end{array}\right) \begin{array}{l} \overline{I} - 2\overline{II} \\ \\ \overline{III} + 7\overline{II} \\ \overline{IV} + 9\overline{II} \end{array} \sim \left(\begin{array}{cccc|c} 1 & 0 & 1 & 5/3 & 11/3 \\ 0 & 1 & 1 & 5/3 & 11/3 \\ 0 & 0 & -2 & -10/3 & -16/3 \\ 0 & 0 & -9 & -19 & -28 \end{array}\right) \overline{III} : (-2) \sim$$

$$\sim \left(\begin{array}{cccc|c} 1 & 0 & 1 & 5/3 & 11/3 \\ 0 & 1 & 1 & 5/3 & 11/3 \\ 0 & 0 & 1 & 5/3 & 8/3 \\ 0 & 0 & -9 & -19 & -28 \end{array}\right) \begin{array}{l} \overline{I} - \overline{III} \\ \overline{II} - \overline{III} \\ \\ \overline{IV} + 9 \end{array} \sim \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 5/3 & 8/3 \\ 0 & 0 & 0 & -4 & -4 \end{array}\right) \overline{IV} : (-4) \sim$$

$$\sim \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 5/3 & 8/3 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right) \overline{III} - \frac{5}{3}\cdot\overline{IV} \sim \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

$x_1 = 1$

$x_2 = 1$

$x_3 = 1$

$x_4 = 1$